

Assignment 2 of MATP4820

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Problem 1

Consider the sequence $\{x^{(k)}\}_{k=0}^{\infty}$ defined by

$$x^{(k)} = \begin{cases} \left(\frac{3}{4}\right)^{2^k}, & \text{if } k \text{ is even} \\ \frac{2}{k+2}x^{(k-1)}, & \text{if } k \text{ is odd} \end{cases}$$

1. Is this sequence Q-quadratically convergent? If yes, prove it; if not, explain why.

No, this sequence is not Q-quadratically convergent.

First, as $k \rightarrow \infty$,

$$\left(\frac{3}{4}\right)^{2^k} \rightarrow 0, \quad \frac{2}{k+2}x^{(k-1)} \rightarrow 0 \implies x^* = 0$$

Now, if k is even:

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2^2} = \frac{\left|\frac{2}{(k+1)+2}x^{(k+1-1)}\right|}{\left|\left(\frac{3}{4}\right)^{2^k}\right|^2} = \frac{\left|\frac{2}{k+3}x^{(k)}\right|}{\left|\left(\frac{3}{4}\right)^{2^k}\right|^2} = \frac{\left|\frac{2}{k+3}\left(\frac{3}{4}\right)^{2^k}\right|}{\left|\left(\frac{3}{4}\right)^{2^k}\right|^2} = \frac{\left|\frac{2}{k+3}\right|}{\left|\left(\frac{3}{4}\right)^{2^k}\right|} = \frac{2}{k+3} \left(\frac{4}{3}\right)^{2^k}$$

Therefore, as $k \rightarrow \infty$

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2^2} = \frac{2}{k+3} \left(\frac{4}{3}\right)^{2^k} \rightarrow \infty$$

Thus, for even k,

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2^2} \rightarrow \infty$$

So, there is no M such that

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2^2} \leq M,$$

Therefore, $\{x^{(k)}\}_{k=0}^{\infty}$ is not Q-quadratically convergent

2. Is this sequence R-quadratically convergent? If yes, prove it; if not, explain why.

Yes the sequence is R-quadratically convergent.

Proof. First, let $\{x^{(k)}\}_{k=0}^{\infty}$ converge to x^* . To show the sequence $x^{(k)}$ is R-quadratically convergent, we have to show there is $\{v^{(k)}\}_{k=0}^{\infty}$ such that

$$\|x^{(k)} - x^*\|_2 \leq v^{(k)}, \text{ and } v^{(k)} \text{ Q-quadratically converges to 0.}$$

Then, as $k \rightarrow \infty$,

$$\left(\frac{3}{4}\right)^{2^k} \rightarrow 0, \text{ and } \frac{2}{k+2}x^{(k-1)} = \frac{2}{k+2} \left(\frac{3}{4}\right)^{2^{(k-1)}} \rightarrow 0$$

Thus, $x^* = 0$. Now, take $\{v^{(k)}\}_{k=0}^{\infty} = \left(\sqrt{\frac{3}{4}}\right)^{2^k}$.

Then if k is even,

$$\|x^{(k)} - x^*\|_2 = \left|\left(\frac{3}{4}\right)^{2^k}\right| \leq \left(\sqrt{\frac{3}{4}}\right)^{2^k} = v^{(k)}$$

If k is odd,

$$\|x^{(k)} - x^*\|_2 = \left|\frac{2}{k+2}x^{(k-1)}\right| = \left|\frac{2}{k+2} \left(\frac{3}{4}\right)^{2^{(k-1)}}\right| \leq \left(\sqrt{\frac{3}{4}}\right)^{2^k} = v^{(k)}$$

Thus, $\|x^{(k)} - x^*\|_2 \leq v^{(k)}$.

Now, we will show $\{v^{(k)}\}_{k=0}^{\infty}$ Q-quadratically converges to 0.

Note, as $k \rightarrow \infty$, $\left(\sqrt{\frac{3}{4}}\right)^{2^k} \rightarrow 0$, so $v^* = 0$, and $v^{(k)}$ converges to 0. Then,

$$\frac{\|v^{(k+1)} - v^*\|_2}{\|v^{(k)} - v^*\|_2^2} = \frac{\left(\sqrt{\frac{3}{4}}\right)^{2^{k+1}}}{\left(\left(\sqrt{\frac{3}{4}}\right)^{2^k}\right)^2} = 1$$

Since the ratio is bounded by 1, the Q-quadratic condition is satisfied, and $v^{(k)}$ Q-quadratically converges to 0. Therefore we have shown there is $\{v^{(k)}\}_{k=0}^{\infty}$ such that $\|x^{(k)} - x^*\|_2 \leq v^{(k)}$, and $v^{(k)}$ Q-quadratically converges to 0, so $\{x^{(k)}\}_{k=0}^{\infty}$ is R-Quadratic. $\square$

Problem 2

Consider the linear system

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ -x_1 + 4x_2 &= 2 \end{aligned} \tag{1}$$

Define the iterative update scheme

$$\begin{aligned} x_1^{(k+1)} &= \frac{x_2^{(k)} + 3}{2}, \\ x_2^{(k+1)} &= \frac{x_1^{(k)} + 2}{4}. \end{aligned} \tag{2}$$

Does the sequence $\{(x_1^{(k)}, x_2^{(k)})\}_{k=0}^{\infty}$ generated by the scheme in (2) from any initial point $(x_1^{(0)}, x_2^{(0)})$ always converge to the solution of the linear system in (3)? If yes, prove it; if not, explain why and give a specific initial point $(x_1^{(0)}, x_2^{(0)})$ such that the generated sequence converges to the solution of (3).

Yes the sequence always converges to the solution of the linear system.

Proof. First, we solve the linear system,

$$\begin{aligned} 2x_1 - x_2 &= 3 \rightarrow x_1 = 2 \\ -x_1 + 4x_2 &= 2 \rightarrow x_2 = 1 \end{aligned} \tag{3}$$

$$\text{So, } x^* = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Then, we can take the error of our current sequence $\{x^{(k)}\}_{k=0}^{\infty} = \{x_1^{(k)}, x_2^{(k)}\}_{k=0}^{\infty}$, to be written as $\|x^{(k)} - x^*\|$.

Now,

$$\|x^{(k)} - x^*\|_2^2 = (x_1^{(k)} - 2)^2 + (x_2^{(k)} - 1)^2$$

Then, after an iteration, the error of our current solution is given by $\|x^{(k+1)} - x^*\|$

$$\|x^{(k+1)} - x^*\|_2^2 = \left\| \begin{bmatrix} \frac{x_2^{(k)} + 3}{2} \\ \frac{x_1^{(k)} + 2}{4} \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} \frac{x_2^{(k)}}{2} - \frac{1}{2} \\ \frac{x_1^{(k)}}{4} - \frac{1}{2} \end{bmatrix} \right\|_2^2 = \frac{(x_2^{(k)} - 1)^2}{4} + \frac{(x_1^{(k)} - 2)^2}{16}$$

Then,

$$\frac{(x_2^{(k)} - 1)^2}{4} + \frac{(x_1^{(k)} - 2)^2}{16} \leq \frac{1}{4} \left((x_1^{(k)} - 2)^2 + (x_2^{(k)} - 1)^2 \right)$$

$$\implies \|x^{(k+1)} - x^*\|_2^2 \leq \frac{1}{4} \|x^{(k)} - x^*\|_2^2 \implies \|x^{(k+1)} - x^*\|_2 \leq \frac{1}{2} \|x^{(k)} - x^*\|_2$$

Now, in general, when iterating

$$\|x^{(k)} - x^*\|_2 \leq \frac{1}{2}^k \|x^{(0)} - x^*\|_2$$

And, as $k \rightarrow \infty$,

$$\frac{1}{2}^k \|x^{(0)} - x^*\|_2 \rightarrow 0.$$

Thus, as we iterate, the error of our solution goes to 0 for any initial point, proving it always converges to the exact solution of the linear system.

□

Problem 3

Let $\{x^{(k)}\}_{k=1}^{\infty}$ and $\{y^{(k)}\}_{k=1}^{\infty}$ be two scalar sequences. Suppose $x^{(k)}$ Q-sublinearly converges to x^* and $y^{(k)}$ Q-linearly converges to y^* .

1. Must $x^{(k)} + y^{(k)}$ be Q-sublinearly convergent to $x^* + y^*$? If yes, prove it; if not, give a counter example.

Yes, $x^{(k)} + y^{(k)}$ must be Q-sublinearly convergent to $x^* + y^*$.

Proof. Assume $\{x^{(k)}\}_{k=1}^{\infty}$ and $\{y^{(k)}\}_{k=1}^{\infty}$ are two scalar sequences, and $\{x^{(k)}\}_{k=1}^{\infty}$

Q-sublinearly converges to x^* , and $\{y^{(k)}\}_{k=1}^{\infty}$ Q-linearly converges y^*

That is, $k \rightarrow \infty$

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2} \rightarrow 1 \text{ and } \frac{\|y^{(k+1)} - y^*\|_2}{\|y^{(k)} - y^*\|_2} \leq r, \text{ for } r \in (0, 1)$$

Now, define $z^{(k)} = x^{(k)} + y^{(k)}$, $z^* = x^* + y^*$. We will show $z^{(k)}$ is Q-sublinearly convergent.

First note, as $k \rightarrow \infty$

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2} \rightarrow 1 \text{ and } \frac{\|y^{(k+1)} - y^*\|_2}{\|y^{(k)} - y^*\|_2} \leq r, \text{ for } r \in (0, 1)$$

This implies $\frac{\|y^{(k)} - y^*\|_2}{\|x^{(k)} - x^*\|_2} \rightarrow 0$, because the Q-sublinear sequence $x^{(k)}$ decays slower than the Q-linear sequence $y^{(k)}$.

Now,

$$\frac{\|z^{(k+1)} - z^*\|_2}{\|z^{(k)} - z^*\|_2} = \frac{|x^{(k+1)} - x^* + y^{(k+1)} - y^*|}{|x^{(k)} - x^* + y^{(k)} - y^*|} = \frac{\left| \frac{x^{(k+1)} - x^*}{x^{(k)} - x^*} + \frac{y^{(k+1)} - y^*}{x^{(k)} - x^*} \right|}{\left| 1 + \frac{y^{(k)} - y^*}{x^{(k)} - x^*} \right|}$$

Now, as we have shown, as $k \rightarrow \infty$,

$$\frac{x^{(k+1)} - x^*}{x^{(k)} - x^*} \rightarrow 1, \quad \frac{y^{(k+1)} - y^*}{x^{(k)} - x^*} \rightarrow 0, \quad \frac{y^{(k)} - y^*}{x^{(k)} - x^*} \rightarrow 0,$$

Thus, as $k \rightarrow \infty$,

$$\frac{\|z^{(k+1)} - z^*\|_2}{\|z^{(k)} - z^*\|_2} = \frac{\left| \frac{x^{(k+1)} - x^*}{x^{(k)} - x^*} + \frac{y^{(k+1)} - y^*}{x^{(k)} - x^*} \right|}{\left| 1 + \frac{y^{(k)} - y^*}{x^{(k)} - x^*} \right|} \rightarrow 1$$

So, $z^{(k)}$ is Q-sublinearly convergent. Since $z^{(k)}$ was defined as $z^{(k)} = x^{(k)} + y^{(k)}$ with $x^{(k)}$ Q-sublinear convergent and $y^{(k)}$ Q-linear convergent, we have proven our claim.

□

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2. Can $x^{(k)} + y^{(k)}$ be Q-linearly convergent to $x^* + y^*$? If yes, give an example; if not, explain why.

No, $x^{(k)} + y^{(k)}$ can never be Q-linearly convergent.

As proven in Part 1, define $z^{(k)} = x^{(k)} + y^{(k)}$ and $z^* = x^* + y^*$. Then

$$\frac{\|z^{(k+1)} - z^*\|_2}{\|z^{(k)} - z^*\|_2} \rightarrow 1 \quad \text{as } k \rightarrow \infty.$$

By definition, a sequence is Q-linearly convergent if there exists $r \in (0, 1)$ such that

$$\frac{\|z^{(k+1)} - z^*\|_2}{\|z^{(k)} - z^*\|_2} \leq r \quad \text{for all sufficiently large } k.$$

Since the limit of the ratio for $z^{(k)}$ is 1, which is not less than 1, it is impossible for the sequence to satisfy the Q-linear convergence criterion. Therefore, $x^{(k)} + y^{(k)}$ cannot be Q-linearly convergent.